

Computer Architecture

Lecture 0. Course Introduction

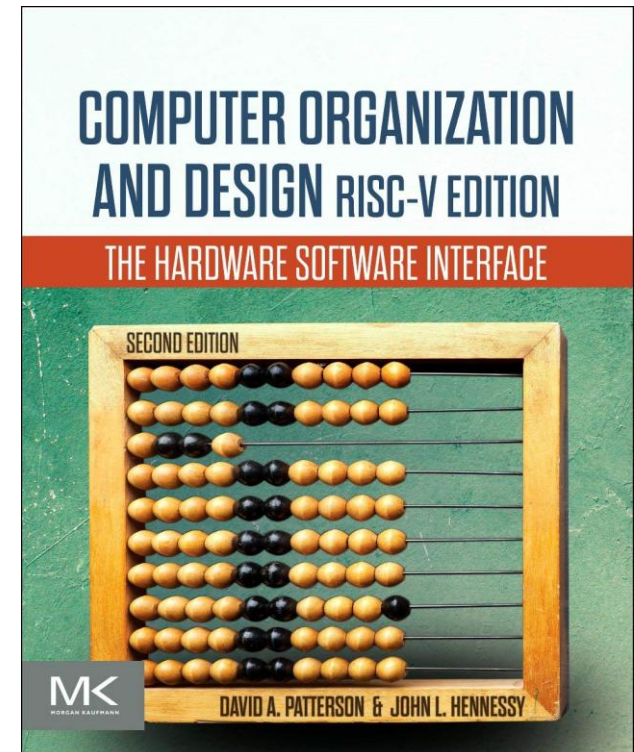
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Computer Engineering
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Acknowledgement: Prof. Taeweon Suh

Course Information

- Instructor
 - Prof. Kon-Woo Kwon
- Textbook
 - **Computer Organization and Design RISC-V Edition**
by David Patterson and John Hennessy, **2nd Ed.**,
Morgan Kaufmann
- Prerequisites
 - **Logic circuits and lab.**
- Contact Information
 - konwoo@hongik.ac.kr

32-bit RISC-V



Your Computers?

- You have spent 2 years as a computer engineering major
- Your computers?



- How is the computer able to execute all those applications?

Abstraction

- Abstraction helps us deal with complexity
 - Hide lower-level detail

Abstraction Analogies

Driver



**Abstraction
layer**



**Machine
Details**



**Combustion
Engine in a car**



**Break system
in a car**

Customer



**Abstraction
layer**



**Machine
Details**



**Hardware board in a
vending machine**

Abstractions in Computer

Programming using APIs



facebook



Operating Systems

Provides APIs (Application Programming Interface)

Instruction Set Architecture (ISA)

Hardware Implementation
(x86)

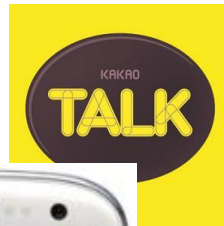


Assembly language
or
Machine language



Abstractions in Smartphones

Application Programming
using APIs



facebook



Operating System



ROID



Instruction Set Architecture
(ISA)

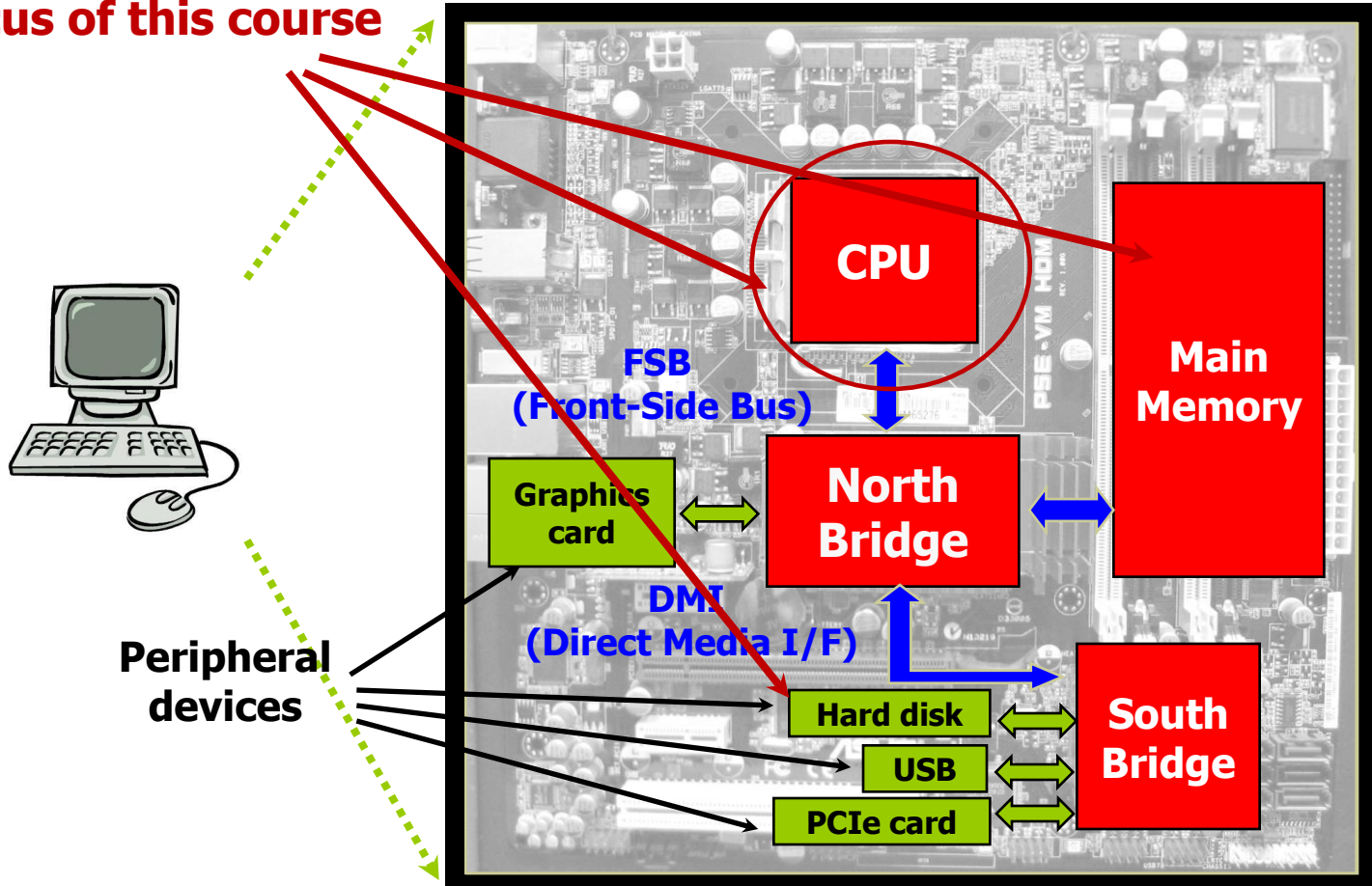


Hardware
Implementation
(ARM)



A Computer System (~ 2010)

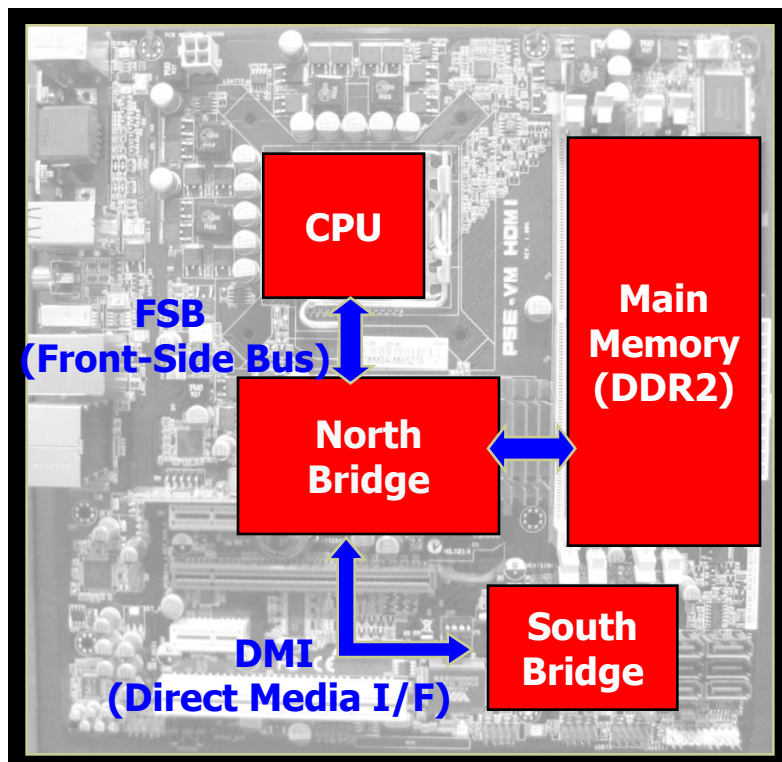
Focus of this course



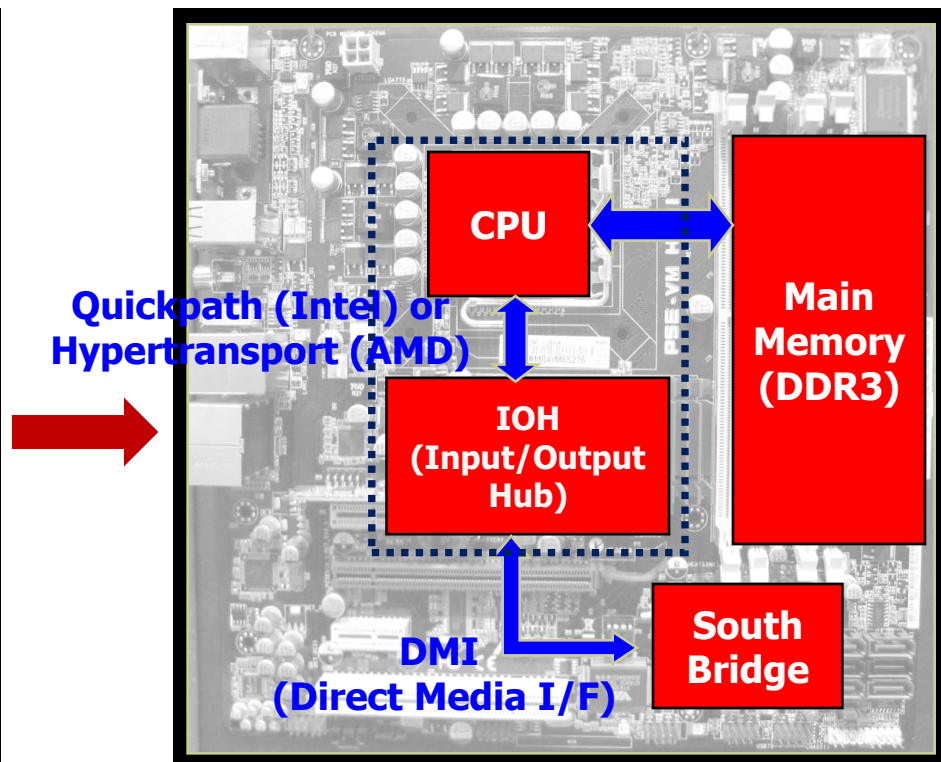
But, don't forget the big picture!

Evolution of Computer Systems

- Core 2 Duo – based Systems

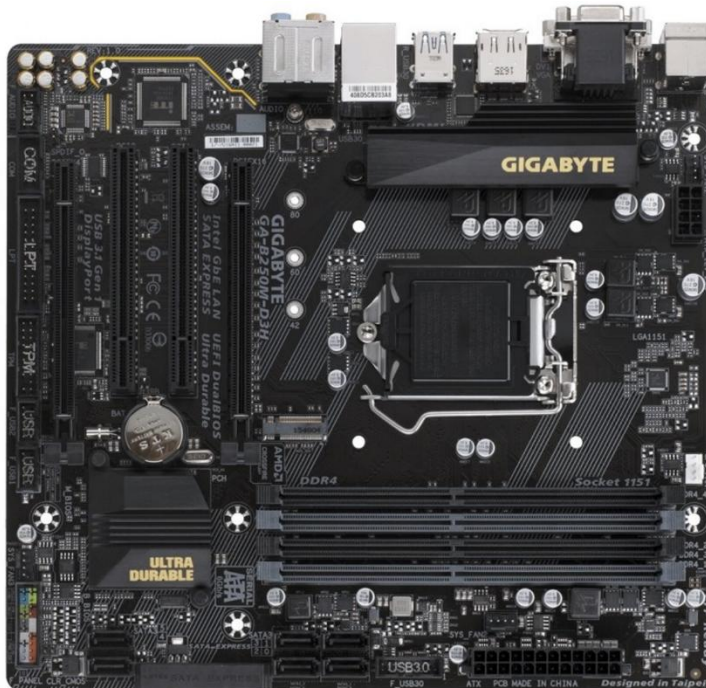


- Core i7– based Systems

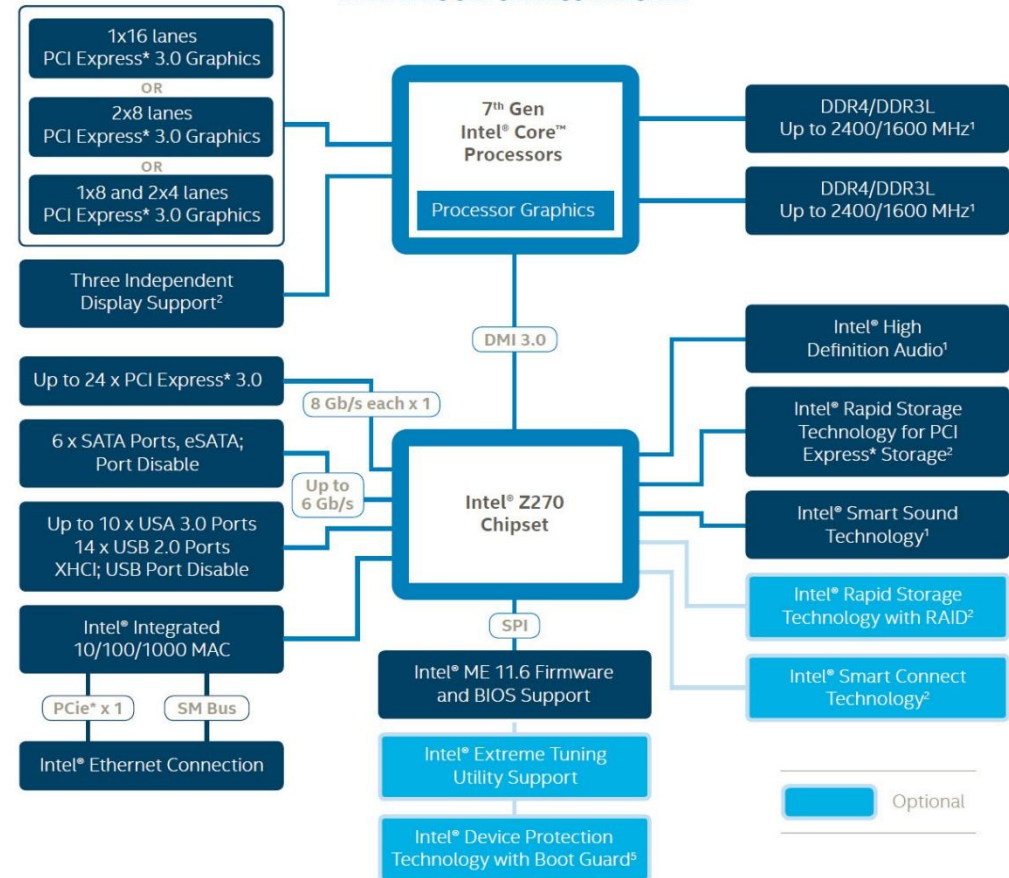


Keep in mind that CPU and computer systems are evolving at a fast pace

7th Gen Core i7 (Kaby Lake)-based Systems



INTEL® Z270 CHIPSET BLOCK DIAGRAM



Ultimate Goal

**Understand How Your Computer
Works Really!**

Digital Logic vs Computer Architecture

- **Logic circuits and lab.**

- We learned digital logic
- What you can do with it?
 - Combinational logic design
 - 7 segment logic, multiplexer etc
 - Sequential logic design
 - Vending machine logic
 - Digital clock logic
 - **But, what you have designed does NOT provide any programmability to users**



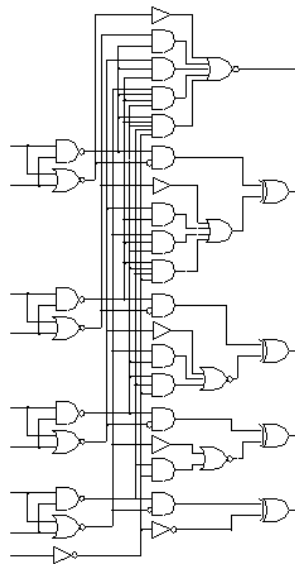
- **Computer Architecture**

- We study and design computers, focusing on CPU (processor)
- **Processor provides programmability to users**

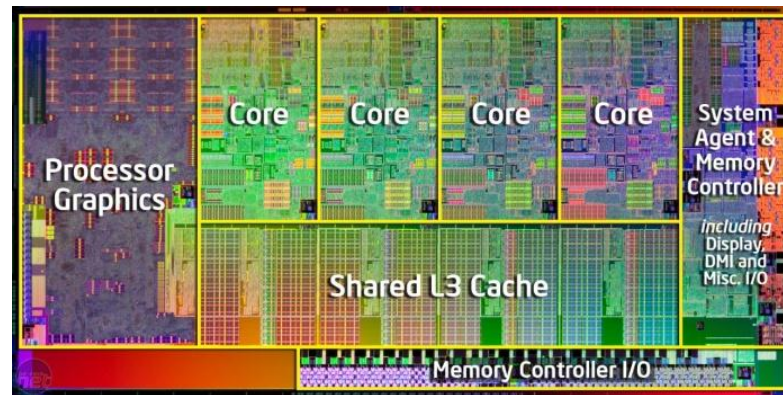


Hardware in Computer

- Computer architecture course mainly covers how to build a CPU with the basic elements

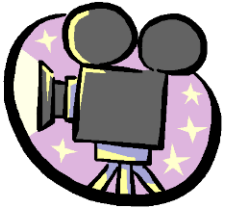


Digital Logic



2nd Generation Core i7 processor
995 million transistors in 216mm² (32nm technology)

Computer Architecture



```
int main()  
{  
    int a, b, c;  
    a = 3;  
    b = 9;  
    c = a + b;  
    return c;  
}
```

- **C Programming**
- **C++, Python ...**



- **Computer Architecture course is where software meets hardware**



- **Digital Logic Design**

Computer Architecture

- Modern x86-based computers are very complicated; It is covered in advanced computer architecture course in graduate school
- Our goal is to learn and understand the basic operation and mechanism of computers, so we focus on
 - New, emerging and simple CPU (32-bit RISC-V) (<https://riscv.org/>)
 - Memory hierarchy (cache, main memory, and hard-disk)

